Duck Nest Survival, Vegetation Density, and Early Weather Conditions in North Dakota

Makena Grigsby

Department of Statistics

STAT 218: Survival Analysis

June 10, 2026

This project examines factors associated with duck nest failure in North Dakota, with a focus on Robel vegetation density measured at nest discovery. Because nests may already be active when they are first found, survival time was revised to begin at estimated nest initiation. Cox proportional hazards models were first fit as a course-aligned comparison, and interval-censored accelerated failure time models were then used as the final modeling approach because many failure times were only known to occur between nest checks. Among the Weibull, log-normal, and log-logistic AFT models, the log-normal model had the lowest AIC. In the final model, higher Robel density was associated with longer nest survival time, warmer first-7-day temperatures were associated with shorter survival time, and first-7-day precipitation was not statistically significant. A Robel density by first-7-day temperature interaction was also examined, but it was not supported after revising the survival time scale, so the simpler additive model was retained.

1 Introduction

Nest survival is an important part of reproductive success in duck populations because failed nests directly reduce the number of young produced during a breeding season. Nest failure can be influenced by several ecological factors, including vegetation structure around the nest, timing within the nesting season, species differences, and weather conditions [1]. In this project, duck nest failure data from North Dakota were analyzed to examine whether local vegetation density was associated with nest survival.

The main predictor of interest was Robel vegetation density, which was measured when the nest was found. Robel values describe visual obstruction from surrounding vegetation, with larger values indicating taller or denser vegetation cover [2]. Greater vegetation obstruction may help conceal nests from predators or reflect habitat conditions that are more favorable for nesting. For this reason, the main goal of the analysis was to estimate the association between Robel density and duck nest survival while adjusting for early weather conditions, species, year, and nest age at discovery.

A key challenge in this analysis was defining the survival time scale. Nests may already be active when they are first discovered, so measuring survival time from the discovery date can miss part of the nesting process. To better match the biology of the data, survival time was measured from the estimated nest initiation date rather than from the first discovery date.

A second challenge was that exact failure times were not always observed. Nests were checked periodically, so many failed nests were known to be active at one check and failed by a later check. This means the true failure time occurred somewhere between two nest checks. Cox proportional hazards models were first fit as a course-aligned comparison, and interval-censored accelerated failure time models were then used as the primary final analysis because they better represented the way failure times were observed [3, 4].

1.1 Data and Variables

The data used in this project come from a field study of monitored duck nests in North Dakota during the 2016 and 2017 nesting seasons [1]. The observational unit was an individual nest. Each nest was followed through repeated field checks until the nest either failed or was last observed without failure. Therefore, the study population for this analysis is monitored duck nests from the North Dakota study area during 2016 and 2017, rather than all duck nests generally.

The event of interest was nest failure, and nests with observed failure were coded as `status = 1`. Nests that were not observed to fail by the final field check were treated as right-censored and coded as `status = 0`. After cleaning the data, one record was removed because it had a nonpositive survival time when time was measured from estimated nest initiation. The cleaned Cox model dataset included 2793 nests, with 1716 observed failures and 1077 censored observations. Of these nests, 1283 were monitored in 2016 and 1510 were monitored in 2017. In 2016, there were 818 observed failures and 465 censored nests. In 2017, there were 898 observed failures and 612 censored nests.

The cleaned dataset represented eight duck species. Gadwall, blue-winged teal, and mallard made up the largest groups, while northern shoveler, northern pintail, lesser scaup, American wigeon, and American green-winged teal were less common.

The main predictor of interest was `robel.found`, the Robel visual obstruction reading recorded when the nest was found. This measurement is taken using a marked Robel pole and represents the height, in centimeters, at which surrounding vegetation visually obstructs the pole [2]. Higher values indicate taller or denser vegetation cover around the nest, which may provide more concealment. In the cleaned dataset, `robel.found` ranged from 2.5 cm to 36 cm, with a median of 13.8 cm and a mean of 14.16 cm. Figure 1 shows that vegetation structure varied noticeably across nests, making Robel density a useful predictor for evaluating whether local nest cover was associated with survival.

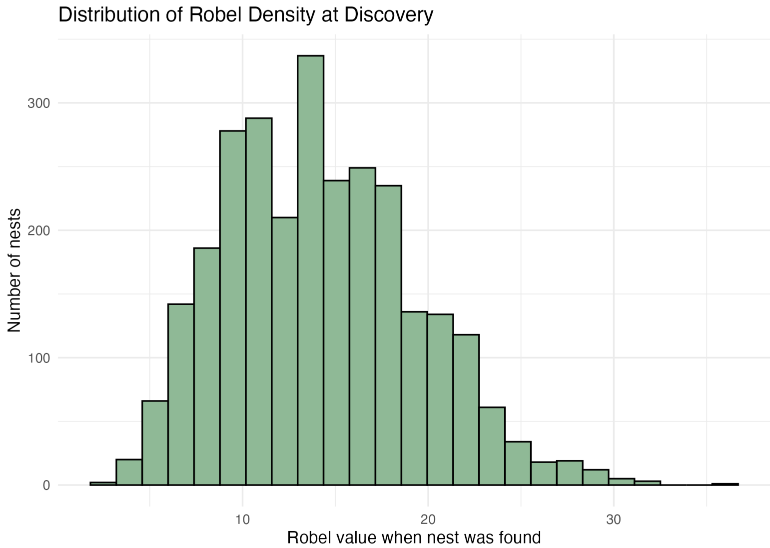

Figure 1: Distribution of Robel vegetation density measured in cm at nest discovery.

Additional variables included study year, species, field habitat, nest vegetation, nest timing variables, and weather summaries. Nest age at discovery was defined as

$$\text{age.found} = \text{FirstFound} - \text{initiation}.$$

This variable was grouped into young, middle, and older age-at-discovery groups. The young group included 1039 nests with ages 1 to 9 days, the middle group included 950 nests with ages 10 to 15 days, and the older group included 804 nests with ages 16 to 57 days. Age at discovery was used as an adjustment variable because nests found at different ages may have different baseline risk patterns. Weather variables were included because early-season weather may influence nest survival. The primary weather covariates were mean temperature and total precipitation during the first 7 days after estimated nest initiation.

1.2 Survival Time Definition

The original analysis measured survival time from the date the nest was first discovered. However, this time scale is not ideal because nests may already be active when they are found. To better represent the biological nesting process, the revised analysis measured survival time from the estimated nest initiation date [3]. This initiation-based time scale is also biologically reasonable because waterfowl incubation commonly lasts about 21 to 31 days, although the exact length varies by species [6].

For the Cox proportional hazards model, each nest needed a single observed follow-up time. Survival time was therefore defined as

$$\text{time} = \text{LastChecked} - \text{initiation}.$$

This definition measures the number of days from estimated nest initiation to the final nest check recorded for that nest. For nests that failed, this treats failure as occurring at `LastChecked`. For nests that did not fail, this represents the last time the nest was observed without failure.

The timing of failure was not always observed exactly. A failed nest could be known to be present at `LastPresent` and then found failed at `LastChecked`. In that case, the true failure time occurred somewhere between those two dates. This creates interval-censored failure times rather than exact failure times. For nests that were not observed to fail, the nest was known to survive through the final check, but the later failure time was not observed, so these nests were treated as right-censored.

Thus, the Cox model used a single initiation-based follow-up time, while the interval-censored AFT model used the lower and upper bounds of the possible failure time. This allowed the final model to better match the way nest failure was observed in the field.

Nest outcome	What was observed	Model time interval
Observed failure	Failed between checks	LastPresent – initiation to LastChecked – initiation
Right-censored	No failure by final check	Begins at LastChecked – initiation; upper time not observed

Table 1: Survival time intervals measured from estimated nest initiation.

1.3 Weather Covariates

Daily weather data were obtained from Daymet using each nest's latitude and longitude coordinates and study year [5]. Because the revised survival time scale began at estimated nest initiation, weather summaries were also calculated beginning at initiation. The main weather variables were mean temperature and total precipitation during the first 7 days after estimated nest initiation.

The first 7 days after initiation were used to define an early weather exposure window that did not depend on the observed survival time. This was important because using weather over the full follow-up period could make the exposure window partly determined by how long the nest survived. Full follow-up weather summaries were also created, but these were treated as exploratory for this reason.

Figure 2 shows the Daymet-based weather information used to create nest-level weather summaries. The plot is included to show how weather covariates were linked to nest locations across the study area.

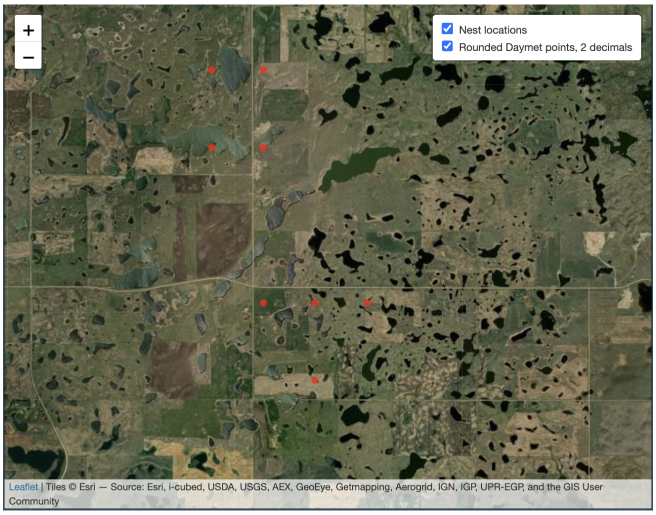

Figure 2: Spatial Daymet weather summaries linked to monitored nest locations in the study area.

Precipitation was transformed using

$$\log(1 + \text{precip.first7}),$$

so that nests with zero precipitation could still be included while reducing the influence of large precipitation values. Mean temperature was kept on its original scale because model diagnostics did not suggest that a more complex form was needed for the main analysis.

1.4 Cox Proportional Hazards Model Comparison

Cox proportional hazards models were first fit using the revised initiation-based survival time [4]. These models were included as a course-aligned comparison and as a way to describe how the covariates were related to the hazard of nest failure [3, 4]. However, the Cox model requires one follow-up time for each nest, so failed nests were treated as failing at the final failed check. This does not fully capture the interval-censored structure of the data, but it provides a useful comparison before fitting the interval-censored models.

The Cox proportional hazards model can be written as

$$h(t \mid Z) = h_0(t)\exp(\beta^\top Z),$$

where $h(t \mid Z)$ is the hazard of observed nest failure at time t, $h_0(t)$ is the baseline hazard, and Z contains nest-level predictors such as vegetation density, early weather, species, year, and age at discovery group.

In this project, year, species, and age at discovery group were handled using stratification. This allowed the baseline hazard to vary across these groups instead of forcing them to share the same baseline risk pattern. The final weather-adjusted Cox model included Robel density, first-7-day mean temperature, and log first-7-day precipitation in the exponential part of the model.

The fitted stratified Cox model was

$$h(t \mid Z_i) = h_{0,s}(t)\exp\left[\beta_1 \text{robel.found}_i + \beta_2 \text{mean.temp.first7}_i + \beta_3 \log(1 + \text{precip.first7}_i)\right],$$

where $h_{0,s}(t)$ is the baseline hazard for stratum s, with strata defined by year, species, and age at discovery group. In this model, covariate effects are interpreted using hazard ratios. A hazard ratio less than 1 indicates a lower hazard of nest failure, while a hazard ratio greater than 1 indicates a higher hazard of nest failure [3, 4].

1.5 Interval-Censored AFT Model

The interval-censored AFT model was used as the primary final model because it better matched how nest failure times were observed [4]. The Cox model was useful as a course-aligned comparison, but it required each nest to have one event or censoring time. For failed nests, the exact failure time was not always known.

The AFT model allowed survival time to remain measured from estimated nest initiation while accounting for the fact that many failures occurred between two nest checks. For failed nests, the lower bound was the number of days from initiation to the last check when the nest was known to be active, and the upper bound was the number of days from initiation to the check when the nest was found failed. For nests that were not observed to fail, the upper failure time was treated as unobserved.

In an accelerated failure time model, covariates are interpreted in terms of survival time rather than the hazard [3, 4]. The model describes whether a covariate is associated with stretching or shortening the expected survival time. In this setting, a time ratio greater than 1 means that the covariate is associated with longer nest survival, while a time ratio less than 1 means that the covariate is associated with shorter nest survival.

A general AFT model can be written as

$$\log(T) = X^\top \beta + \sigma\varepsilon,$$

where T is the survival time, $X^\top\beta$ contains the covariate effects, σ is a scale parameter, and ε follows the distribution for the chosen AFT model. Positive coefficients correspond to longer survival time, while negative coefficients correspond to shorter survival time. Exponentiating the coefficients gives time ratios.

Several common interval-censored AFT distributions were considered, and the final distribution was selected using AIC [4]. Among the Weibull, log-normal, and log-logistic AFT models, the log-normal model had the lowest AIC and was selected as the final interval-censored model.

The final log-normal interval-censored AFT model was

$$\begin{aligned}\log(T_i) = \beta_0 &+ \beta_1 \text{robel.found}_i + \beta_2 \text{mean.temp.first7}_i \\ &+ \beta_3 \log(1 + \text{precip.first7}_i) + \gamma_{\text{year}(i)} \\ &+ \alpha_{\text{age group}(i)} + \delta_{\text{species}(i)} + \sigma\varepsilon_i.\end{aligned}$$

where T_i is the initiation-based survival time for nest i. The coefficients β_1, β_2, and β_3 describe the effects of Robel density, first-7-day mean temperature, and log first-7-day precipitation on log survival time. The terms $\gamma_{\text{year}(i)}$, $\alpha_{\text{age group}(i)}$, and $\delta_{\text{species}(i)}$ represent adjustment effects for the year, age-at-discovery group, and species of nest i. The error term ε_i follows the log-normal AFT model assumption, and σ is the scale parameter.

In this model, positive coefficients correspond to longer survival time and negative coefficients correspond to shorter survival time. Exponentiating each coefficient gives a time ratio, which was used to interpret the final AFT results. The model was fit using initiation-based failure-time intervals for failed nests and right-censoring after the final check for nests that were not observed to fail.

1.6 Model Diagnostics and Sensitivity Checks

For the Cox model, the proportional hazards assumption was assessed using Schoenfeld residuals and the global Schoenfeld test. Cox-Snell residuals were used as an overall model fit check. For the interval-censored AFT model, candidate distributions were compared using AIC, and additional residual checks were used informally because many failure times were interval-censored.

Sensitivity analyses were performed to check whether the main conclusion depended on specific modeling choices. These included a midpoint timing model for failed nests, removal of nests with age at discovery greater than 35 days, spline terms for weather variables, and a Robel density by first-7-day temperature interaction.

2 Results

2.1 Exploratory Survival Results

After redefining survival time to begin at estimated nest initiation, the cleaned dataset contained 2793 nests. There were 1716 observed failures and 1077 censored observations. The initiation-based survival times ranged from 5 to 58 days, with a median of 29 days.

Kaplan-Meier curves and log-rank tests were used as unadjusted comparisons of survival across categorical groups. These results suggested differences in nest survival by year, Robel density group, species, and age at discovery group. Field habitat group and nest vegetation group did not show strong evidence of different survival curves.

The separation by age at discovery group was expected because age at discovery is closely tied to the initiation-based time scale. For this reason, age at discovery group was treated as a timing-related adjustment variable rather than a primary ecological predictor.

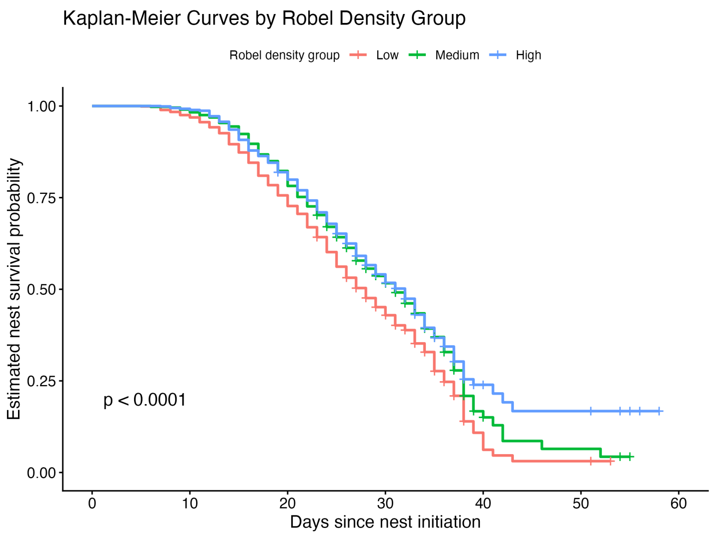

Figure 3: Kaplan-Meier survival curves by Robel density group using initiation-based survival time.

The Kaplan-Meier curves for Robel density group in Figure 3 were consistent with the main research question. Nests in higher Robel density groups tended to have higher estimated survival probabilities than nests in lower Robel density groups. Since these curves do not adjust for other variables, they were used as descriptive evidence and motivation for the regression models.

2.2 Cox Model Results

A Cox proportional hazards model was first fit using the initiation-based survival time. This model was used as a course-aligned comparison because it treats each nest as having one event or censoring time. The final weather-adjusted Cox model included Robel density, first-7-day mean temperature, and log first-7-day precipitation, with stratification by year, species, and age at discovery group. The main Cox model estimates are shown in Table 2.

Variable	Hazard Ratio	Lower 95%	Upper 95%	p-value
Robel density	0.9813	0.9694	0.9934	0.0025
First-7-day temperature	1.0189	1.0022	1.0359	0.0267
Log first-7-day precipitation	1.0249	0.9857	1.0657	0.2177

Table 2: Key results from the final weather-adjusted stratified Cox model.

The Cox model gave the same general direction as the exploratory survival curves. Higher Robel density was associated with a lower hazard of nest failure. The hazard ratio for `robel.found` was 0.9813, meaning that each one-unit increase in Robel density was associated with about a 1.9% lower hazard of failure, holding the weather variables fixed within the stratified model.

First-7-day temperature was also associated with nest failure. The hazard ratio for `mean.temp.first7` was 1.0189, suggesting that warmer early conditions were associated with a higher failure hazard. Log first-7-day precipitation was not statistically significant in this model.

Because one-unit changes can be small, the Cox model estimates were also scaled to larger changes in Robel density and first-7-day temperature. These scaled interpretations are shown in Table 3.

Predictor	Change	Hazard ratio	Interpretation
Robel density	5 cm increase	0.91	9% lower failure hazard
Robel density	10 cm increase	0.83	17% lower failure hazard
First-7-day temperature	5°C increase	1.10	10% higher failure hazard
First-7-day temperature	10°C increase	1.21	21% higher failure hazard

Table 3: Scaled Cox model interpretations for larger changes in Robel density and first-7-day temperature.

A Robel density by first-7-day temperature interaction was also checked because an earlier discovery-based version of the analysis suggested that the Robel association might change with temperature. In the revised initiation-based Cox model, the interaction was not supported. The interaction term was not statistically significant ($p = 0.510$), and the likelihood ratio test also did not support adding it ($p = 0.509$). For this reason, the simpler additive Cox model was retained.

After stratifying by year, species, and age at discovery group, the proportional hazards assumption appeared reasonable. The global Schoenfeld residual test for the final weather-adjusted Cox model had a p-value of 0.61, so there was not strong evidence against the proportional hazards assumption for the modeled covariates.

2.3 Interval-Censored AFT Model Results

The Cox model was useful as a course-aligned comparison, but it required each nest to have one event or censoring time. Because many failures were only known to occur between two nest checks, the interval-censored AFT model was used as the final modeling approach. This allowed survival time to remain measured from estimated nest initiation while accounting for uncertainty in the exact failure time.

For failed nests, the exact failure date was not always known. Instead, the failure time was represented using initiation-based lower and upper bounds. The lower bound was defined as the number of days from estimated initiation to the last check when the nest was known to be active, and the upper bound was defined as the number of days from estimated initiation to the check when the nest was found failed. For nests that were not observed to fail, the upper failure time was treated as unobserved, so these nests were right-censored after their final check.

The interval-censored dataset included 2788 nests, with 1711 failures and 1077 censored observations. Five failed nests were excluded because the lower and upper failure-time bounds were equal, producing a zero-length failure interval.

Several interval-censored AFT distributions were compared using AIC. Lower AIC values indicate better relative model fit after accounting for model complexity. As shown in Table 4, the log-normal model had the lowest AIC and was selected as the final interval-censored AFT model.

Table 4: AIC comparison for interval-censored AFT models.

Model	AIC
Weibull AFT	8199.165
Log-normal AFT	7914.141
Log-logistic AFT	7987.101

Table 5: Key results from the final log-normal interval-censored AFT model.

Variable	Time Ratio	Lower 95%	Upper 95%	p-value
Robel density	1.0090	1.0038	1.0142	0.00076
First-7-day temperature	0.9920	0.9848	0.9992	0.0297
Log first-7-day precipitation	0.9911	0.9739	1.0086	0.3170

Robel density was associated with longer nest survival time. The estimated time ratio for `robel.found` was 1.0090, meaning that each one-unit increase in Robel density was associated with about a 0.9% longer survival time, holding the other variables fixed. First-7-day temperature was associated with shorter survival time, while log first-7-day precipitation was not statistically significant.

Because one-unit changes can be small, the AFT model estimates were also scaled to larger changes in Robel density and first-7-day temperature. These scaled interpretations are shown in Table 6.

Predictor	Change	Time ratio	Interpretation
Robel density	5 cm increase	1.05	5% longer survival time
Robel density	10 cm increase	1.09	9% longer survival time
First-7-day temperature	5°C increase	0.96	4% shorter survival time
First-7-day temperature	10°C increase	0.92	8% shorter survival time

Table 6: Scaled AFT model interpretations for larger changes in Robel density and first-7-day temperature.

These scaled estimates make the AFT results easier to interpret in practical units, since 5 cm and 10 cm changes in Robel density and 5°C and 10°C changes in temperature are more meaningful than one-unit changes alone.

2.4 Diagnostics and Sensitivity Checks

For the Cox model, the proportional hazards assumption was checked using Schoenfeld residuals. After stratifying by year, species, and age at discovery group, the global proportional hazards test was not statistically significant, so there was not strong evidence that the modeled covariate effects changed over time.

The Cox-Snell residual plot in Figure 4 suggested that the Cox model fit the data reasonably well through most of the residual range, but the upward deviation in the upper tail indicated that the model did not fully capture the largest residual values. This was consistent with the role of the Cox model as a useful comparison model rather than the primary final model, since the Cox model required assigning each failed nest a single event time.

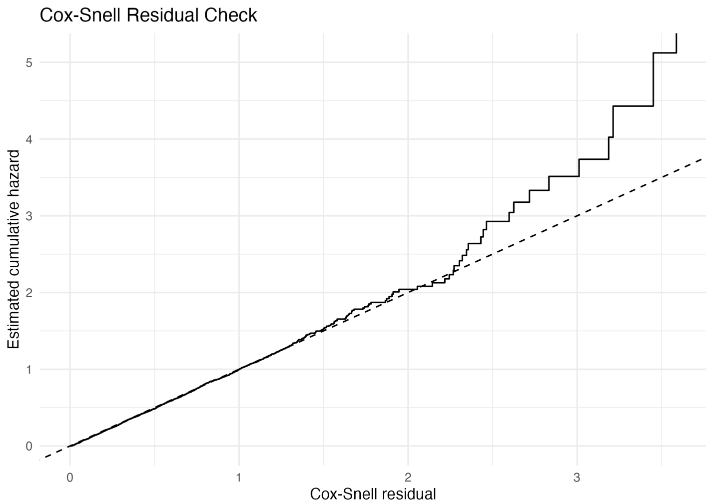

Figure 4: Cox-Snell residual check for the stratified Cox model. The curve follows the reference line reasonably well through most of the residual range, with some deviation in the upper tail.

Check	Result	Decision
Schoenfeld residual test	Global $p = 0.61$	PH assumption reasonable
Temperature spline, Cox	Linear AIC = 13188.03; spline AIC = 13191.49	Keep linear temperature
Temperature spline, AFT	Linear AIC = 7914.141; spline AIC = 7917.803	Keep linear temperature
Temperature KM groups	Log-rank $p = 0.30$	No strong unadjusted group difference
Robel × temperature, Cox	Interaction $p = 0.510$; LRT $p = 0.509$	Keep additive model
Robel × temperature, AFT	Additive AIC = 13423.69; interaction AIC = 13424.10	Keep additive model
Midpoint timing	Robel HR = 0.9815	Main conclusion stable
Age ≤ 35 sensitivity	Robel HR = 0.9819	Main conclusion stable
Precipitation spline	Robel HR = 0.9812	Main conclusion stable

Table 7: Summary of diagnostic and sensitivity checks.

Overall, the diagnostics and sensitivity checks supported the simpler additive modeling approach. The temperature spline models did not improve fit compared with the linear temperature models, and the Robel density by first-7-day temperature interaction was not supported in either the Cox model or the interval-censored AFT model. Timing, age-at-discovery, and precipitation spline sensitivity checks gave similar Robel density conclusions, suggesting that the main finding was not driven by one specific modeling choice.

3 Discussion

This analysis found evidence that Robel vegetation density was associated with duck nest survival in North Dakota. In the Cox model, higher Robel density was associated with a lower hazard of nest failure, while in the interval-censored log-normal AFT model, higher Robel density was associated with longer survival time. Although these two models use different interpretations, they supported the same main conclusion: nests with greater vegetation obstruction tended to have better survival outcomes. Early temperature was also related to survival. Higher first-7-day temperature was associated with a higher failure hazard in the Cox model and shorter survival time in the AFT model. In contrast, first-7-day precipitation was not statistically significant in either model, so it was treated as an adjustment variable rather than a main finding.

A major part of the final analysis was redefining survival time from estimated nest initiation instead of nest discovery. This better matched the biology of the nesting process because nests may already be active when they are first found. The interval-censored AFT model also better matched the field data because many failures were only known to occur between two nest checks. Among the candidate AFT models, the log-normal model had the lowest AIC and was selected as the final model. Sensitivity checks supported the main Robel density finding. The conclusion remained similar after using midpoint timing for failed nests, removing nests with high age at discovery, and allowing weather effects to enter the model more flexibly. The Robel density by temperature interaction was also checked, but it was not supported after revising the time scale.

This analysis does have some limitations. The data are observational, exact failure times were not always observed, and some initiation dates may have estimation error. In addition, Daymet weather summaries are gridded estimates and may not fully capture nest-level microclimate conditions. Future work could build on this analysis by using more detailed nest-level data. More frequent nest checks would reduce uncertainty in failure timing, while direct microclimate measurements could better represent the weather conditions experienced at each nest. Additional habitat, predator, or spatial information could also help explain variation in nest failure that was not captured by the current models.

Overall, the results suggest that local vegetation density was positively associated with duck nest survival. Nests with higher Robel visual obstruction readings tended to have lower failure hazard and longer survival time, while warmer first-7-day temperatures were associated with poorer survival outcomes. This temperature finding may be important in the context of warming seasonal conditions, since higher early temperatures could place additional stress on nesting success if similar patterns continue over time. These findings support the importance of considering both local nest structure and early weather conditions when studying duck nest survival.

4 Acknowledgments

I thank Dr. Vivian Li for the course instruction and feedback throughout STAT 218. I also acknowledge Ringelman and Skaggs for collecting and making the duck nest survival data available for use in this analysis. AI assistance was used for some code troubleshooting when encountering errors. However, all final decisions with the code, analyses, and project materials are my own. All of which are available on GitHub.

References

[1] Ringelman, K. M., and Skaggs, C. G. (2019). Vegetation phenology and nest survival: Diagnosing heterogeneous effects through time. *Ecology and Evolution*, 9, 2121–2130. https://doi.org/10.1002/ece3.4906

[2] Robel, R. J., Briggs, A. D., Dayton, A. D., and Hulbert, L. C. (1970). Relationship between visual obstruction measurements and weight of grassland vegetation. *Journal of Range Management*, 23, 295–297.

[3] Schober, P., and Vetter, T. R. (2018). Survival analysis and interpretation of time-to-event data: The tortoise and the hare. *Anesthesia & Analgesia*, 127(3), 792–798. https://doi.org/10.1213/ANE.0000000000003653

[4] Li, Vivian Wei. (2026). *STAT 218 Lecture Notes*. Department of Statistics, University of California, Riverside. Unpublished course materials.

[5] Thornton, M. M., Shrestha, R., Wei, Y., Thornton, P. E., Kao, S.-C., and Wilson, B. E. (2022). Daymet: Daily Surface Weather Data on a 1-km Grid for North America, Version 4 R1. ORNL DAAC, Oak Ridge, Tennessee, USA. https://doi.org/10.3334/ORNLDAAC/2129

[6] Wells-Berlin, A. M., Dzus, E. H., and Arnold, T. W. (2005). Incubation length of dabbling ducks. *The Condor*, 107(3), 700–704. https://pubs.usgs.gov/publication/70027617